
Feature Learning in Supervised AI Models

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Abstract

1 In this research paper we will learn and investigate how various supervised learning
2 models learn features. We will do this through four tasks. In task 1 we will see
3 how strong inductive bias is in models and how their architecture can effect how
4 they learn features (i.e. whether they learn shape or texture or some other attribute).
5 In task 2 we will observe how this effects Domain Generalization. The idea being
6 that we want the model to learn identifying features about the objects (like their
7 shape) instead of Domain Specific features (like if the dataset had grass behind
8 every cat and the model associated the grass with the cat). If the model can do this,
9 it should perform well even on Domain Shifts. We extend this idea in task 3 with
10 Domain Generalization and completely unknown target domains. Finally, in task 4
11 we tackle the open set problem, to see if we can identify casual features even on
12 unknown classes, and separate them as they come.

13 1 Introduction

14 We will do various experiments in this research paper to study how different AI models learn features,
15 and the consequence of their architectures. This is done in four parts.

16 First, we will see the study the influence of Inductive Biases and Feature Representations, our goal
17 here is to see the influence of model architecture. By comparing the metrics of four well known and
18 standard models (ResNet-50, ViT-B, CLIP). We will first perform a baseline metric on all of them,
19 and then perform 5 different experiments. Colour Bias: We perform two different types, grey scale,
20 and hue shift, and see how well the models perform. Shape vs Texture (cue-conflict): We apply the
21 texture of one object to another, to see if models are using texture or shape for classification and
22 which is more important. Translation: We shift the images, left, right, up, or down and see if the
23 model can still classify. Lastly, we shuffle the patches of the images to see how the global structure
24 effects the models.

25 Our points of interest are, what cue-conflict and colour reveal about reliance on shape, texture, and
26 colour, and what coverage can tell us about shape bias. How the models respond when spatial structure
27 is altered and what that tells us about the importance of locality, global organisation, and positional
28 sensitivity. To what extent prediction-level changes agree with changes in feature representations,
29 and which observed difference can reasonably be attributed to architecture as opposed to pretraining
30 data, supervision, augmentation or model capacity.

31 2 Experimental Setup/Methods

32 2.1 Inductive Biases and Feature Representations

33 Models:

34 • torchvision ResNet-50 with ResNet50_Weights.IMAGENET1K_V2;

- torchvision ViT-B/16 with ViT_B_16_Weights.IMAGENET1K_V1;
- OpenCLIP ViT-B-32 with pretrained='openai'.

Dataset: STL-10

Configuration:

- Seed: 6304
- Optimizer: AdamW
- Learning Rate: 10^{-3}
- Weight Decay: 10^{-4}

For the Shape vs Texture we used AdaIN implementation, which follows Huang & Belongie (2017). the encoder/decoder architecture and the pretrained weights come from the public reimplementation naoto0804/pytorch-AdaIN, weights mirrored at BhupatiNadar/adain-style-transfer-models on HuggingFace.

3 Results

3.1 Inductive Biases and Feature Representations

We first take some basic measurements as a baseline, we freeze the three classifier heads and measure top-1 accuracy, macro-F1, and mean maximum confidence. We also do this for zero shot CLIP, we measure confidence from the softmax over scaled class similarities.

Table 1: Clean Baseline - Shows all models performing very well on the dataset.

System	Top-1 (%)	Macro-F1 (%)	Mean max conf.
ResNet-50 + linear head	96.4	96.39	0.91
ViT-B/16 + linear head	97.2	97.21	0.97
CLIP ViT-B-32 + linear head	96.6	96.60	0.27
CLIP ViT-B-32 zero-shot	97.0	97.01	0.94

All the models perform well as a baseline, this maybe a limitation of the STL-10 dataset, as its distinct classes would be easier to classify. This can be a limitation as differences in experiments would more visibly show a decrease rather than an increase. It should also be noted that the CLIP Max Mean is low because it uses Cosine similarity in training, and the L-2 norm forces the magnitude to be 1. So when attached with a linear head and no compensation, the value comes off as low. It does not mean it is lower in confidence than other models, just that rows can't be compared the magnitude lower, not because it is less confident than others

3.1.1 Color Bias

We did two experiments with colour, the first we changed the images to greyscale measured the same metrics and got the following results:

Table 2: Greyscale: Minimal change in Top-1 Accuracy

System	Top-1	Macro-F1	Δ Top-1	Consistency	Conf.
ResNet-50 + linear head	94.6	94.61	-1.8	95.8	0.91
ViT-B/16 + linear head	95.4	95.41	-1.8	98.0	0.96
CLIP ViT-B-32 + linear head	93.8	93.78	-2.8	93.0	0.25
CLIP ViT-B-32 zero-shot	93.0	92.97	-4.0	94.0	0.93

We can see that making the image greyscale results in a slight decrease in accuracy throughout. Given that we have removed information from the image this is to be expected. A more interesting

64 observation is that there was no significant difference between Convolution and Vision Transformers,
65 though consistency is slightly higher for the Vision Transformer. CLIP had the most sever impact.
66 Both of the readings are lower than either ResNet or ViT-B, and further zero-shot is the lowest.
67 One plausible explanation for this is that the text embeddings had an influence on how accurately
68 important colour is during pre training. For example if a prompt were to be "A picture of a Brown
69 Dog." that would influenced both the visual encoder and the text encoder. It would also explain why
70 zero-shot performed even worse. This though is not conclusive, and would be hard to confirm.

71 The second experiment was with a Hue Shift of 40 deg. A limitation of this is that a hue shift will only
72 have a good effect on well saturated images. On less saturated images the change is very minimal,
73 making the result less significant.

Table 3: Hue Rotation of 40: Minimal change in accuracy

System	Top-1	Macro-F1	Δ Top-1	Consistency	Conf.
ResNet-50 + linear head	94.4	94.42	-2.0	95.2	0.90
ViT-B/16 + linear head	95.8	95.79	-1.4	98.2	0.96
CLIP ViT-B-32 + linear head	94.8	94.78	-1.8	95.8	0.26
CLIP ViT-B-32 zero-shot	95.2	95.18	-1.8	95.6	0.93

74 Again we see a minimal change in accuracy, this time they are much closer to each other. No claims
75 can be made as the difference is small enough that it can be considered noise. Again though, we will
76 find that ViT is just slightly more consistent than other models. It is also of note that unlike greyscale,
77 CLIP is much more in line with other models. One explanation for this is that in greyscale, we totally
78 lost colour information, however with a rotation, the relationship between the colour of the objects is
79 still the same, it has just been rotated. Hence, the loss is less with CLIP now.

80 3.1.2 Shape vs. Texture

81 For this comparison alpha strength of 1 was required, as lower values did not transfer texture well
82 enough. We get the following results:

83 3.2 Cross-Task Discussion

84 4 Conclusions

85 5 References